

Nikhil Talagana

(Database Administrator)

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Summary

Database Administrator with 2+ years of experience in PostgreSQL, Oracle, and AWS RDS/Aurora. Strong background in automating DBA operations using Python and AWS services. Actively building AI-assisted tools using Lang Chain and RAG to enhance database monitoring, deployment validation, and incident resolution.

Skills

Programming Languages	: Python, Java
Database	: PostgreSQL, Oracle
Operating System	: Linux
Cloud (AWS)	: Aurora RDS, EC2, S3, IAM, CloudWatch, SSM, Lambda, CloudTrail, Bedrock
Tools	: Git, DBeaver, PL/SQL Developer
GenAI	: RAG, LLM, Lang Chain, Lang Graph

Professional Experience

Database Administrator Vitech Systems Asia Pvt Limited	March 2024 – Present Hyderabad, Telangana
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❖ PostgreSQL & AWS Aurora RDS DBA

- Installed and configured PostgreSQL software on Linux and Windows servers.
- Created and maintained PostgreSQL databases.
- Configured and managed pg_hba.conf for authentication and access control.
- Performed backup and restore operations using pg_dump and pg_restore.
- Created and managed database templates and clones for environment refreshes.
- Performed schema-level, metadata backups and restores using pg_dump and pg_restore.
- Performed table-level backups and restores for selective data recovery and troubleshooting.
- Created and administered AWS Aurora PostgreSQL RDS clusters.
- Performed Aurora RDS cluster cloning for non-production environments.
- Created RDS snapshots and performed cross-account snapshot copy.
- Restored databases from RDS snapshots and performed Point-in-Time Recovery (PITR).
- Worked on Aurora RDS patching and version upgrades.
- Worked on scale up and scale down activities based on client requirements.
- Generated pgBadger reports
- Performed routine VACUUM and ANALYZE operations across entire databases as well as specific tables based on performance requirements.
- Used pg_repack to reorganize tables and indexes with minimal locking and downtime.
- Created and managed database users and roles, including read-only and application-specific access and assigned and maintained schema-level and object-level privileges.
- Performed RDS scale up and scale down activities based on request.
- Performed data scrambling/masking in PostgreSQL databases to secure sensitive data during lower-environment refreshes.

❖ Oracle Database Administration

- Understanding of Oracle Memory Architecture (SGA, PGA).
- Installed Oracle software on Linux and Windows platforms.
- Created Oracle databases using DBCA (GUI) and silent mode with response files.
- Configured and maintained Oracle Listener and TNS networking.
- Configured and managed Undo Tablespaces.
- Performed Oracle database defragmentation activities.
- Created and managed ASM disk partitions.
- Performed RMAN backup and restore operations.
- Executed logical backups and restores using EXPDP / IMPDP utilities.
- Created and managed database users and roles, including read-only and application-specific access.
- Assigned and maintained schema-level and object-level privileges.
- Performed data scrambling/masking in Oracle databases to secure sensitive data during lower environment refreshes.
- Exposure to Oracle Multitenant architecture (CDB / PDB).
- Exposure to Oracle RAC environments, including monitoring and basic administrative support.

Automation & Enhancements:

- Automated Blue-Green deployment creation, including Green RDS provisioning and switchover, to enable low-downtime PostgreSQL releases.
- Implemented cross-account RDS snapshot copy between AWS accounts for DR and environment refreshes.
- Enhanced pg_restore automation to dynamically calculate parallel restore jobs based on RDS instance class and database count.
- Developed a Python-based monitoring script to capture PostgreSQL lock details every minute for proactive troubleshooting.

GenAI Initiatives:

- Built an AI-assisted validation framework to analyze Blue-Green deployment packages and ensure database compatibility prior to deployment.
- Developed an AI-powered Jira Bot to suggest solutions for newly created Jira tickets.
- Generated weekly AI-based metrics comparing developer-implemented fixes versus AI-suggested solutions.
- Automated weekly ingestion of closed Jira tickets into a vector database to continuously improve AI-assisted analysis and recommendations.

Education

Bachelor of Technology (EEE)

University College of Engineering Narasaraopet, JNTUK.

CGPA: 6.68/10

2019-2023

AP, India

Professional Training

- **AWS Cloud Computing** – Gana Technologies, Hyderabad
(Hands-on training on EC2, RDS, S3, IAM, CloudWatch, VPC basics)